SAMRAT UTSAV

TECHNOVISION 2024

WELCOME

DEPARTMENT OF ELECTRICAL ENGINEERING

TOPIC - SMART ELECTRIC DISTRIBUTION BOX

BY NIKIHIL KUSHWAH AND SURYANSH VAIDYA

B.TECH 1ST YEAR (E.E.)

OUR MENTOR - DR. MONIKA JAIN

THE TIMES OF INDIA

Over 1 crore consumers never paid their power bills in UP

Pankaj Shah & Arvind Chauhan / TNN /
Updated: Oct 23, 2020, 14:11 IST

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Out of 5.5 lakh power consumers in Bhopal, 30% don't pay bill every month: Study

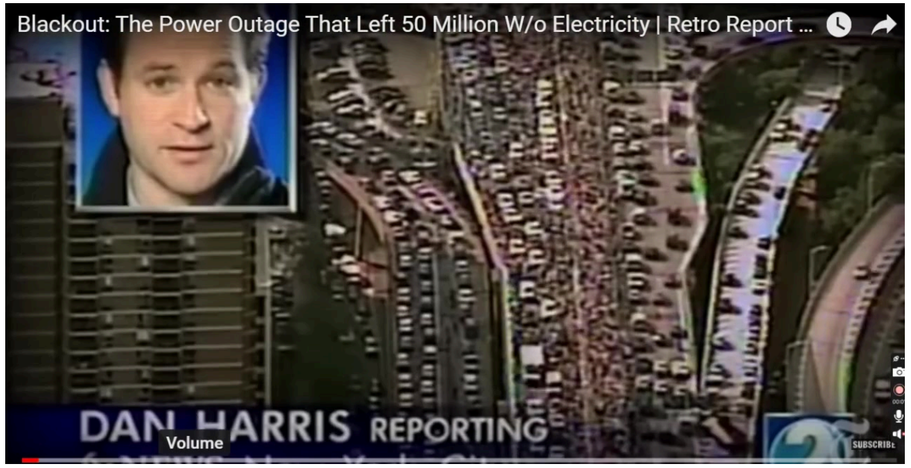
TNN / Updated: Apr 30, 2023, 09:05 IST

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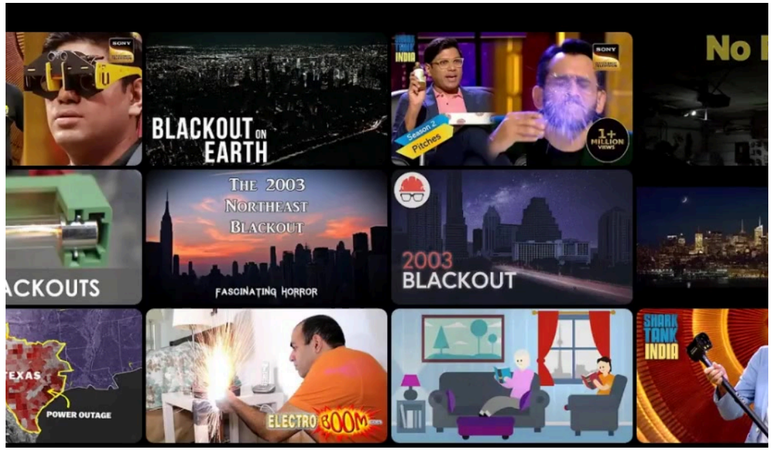
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According to the official analysis of the blackout prepared by the US and Canadian governments, more than 508 generating units at 265 power plants shut down during the outage. In the minutes before the event, the NYISO-managed power system was carrying 28,700 MW of load. At the height of the outage, the load had dropped to 5,716 MW, a loss of 80%.[6]

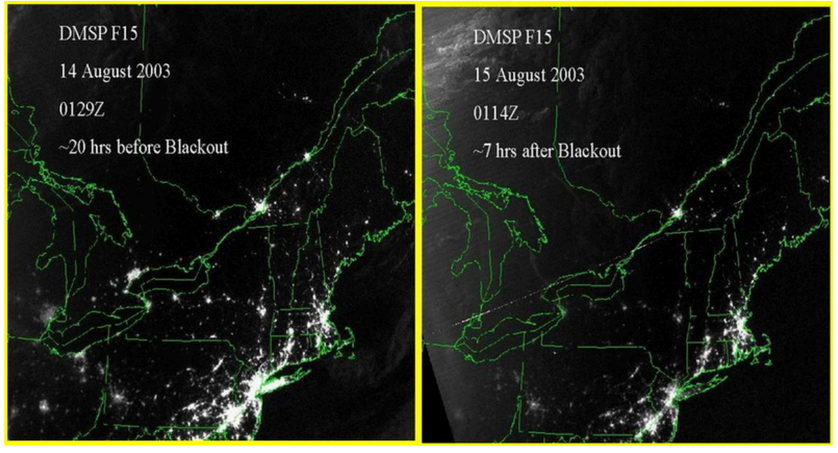
Essential services remained in operation in some of these areas. In others, backup generation systems failed. Telephone networks generally remained operational, but the increased demand triggered by the blackout left many circuits overloaded. Water systems in several cities lost pressure, forcing boil-water advisories to be put into effect. Cellular service was interrupted as mobile networks were overloaded with the increase in volume of calls.[7] Many cell sites were out of commission due to power outages.[8] Television and radio stations remained on the air, with the help of backup generators, although some stations were knocked off the air for periods ranging from several hours to the length of the entire blackout. [*vague*]



It was a hot day (over 31 °C, or 88 °F) in much of the affected region, and the heat played a role in the initial event that triggered the wider power outage. The high ambient temperature increased energy demand, as people across the region turned on fans and air conditioning. This caused the power lines to sag as higher currents heated the lines.





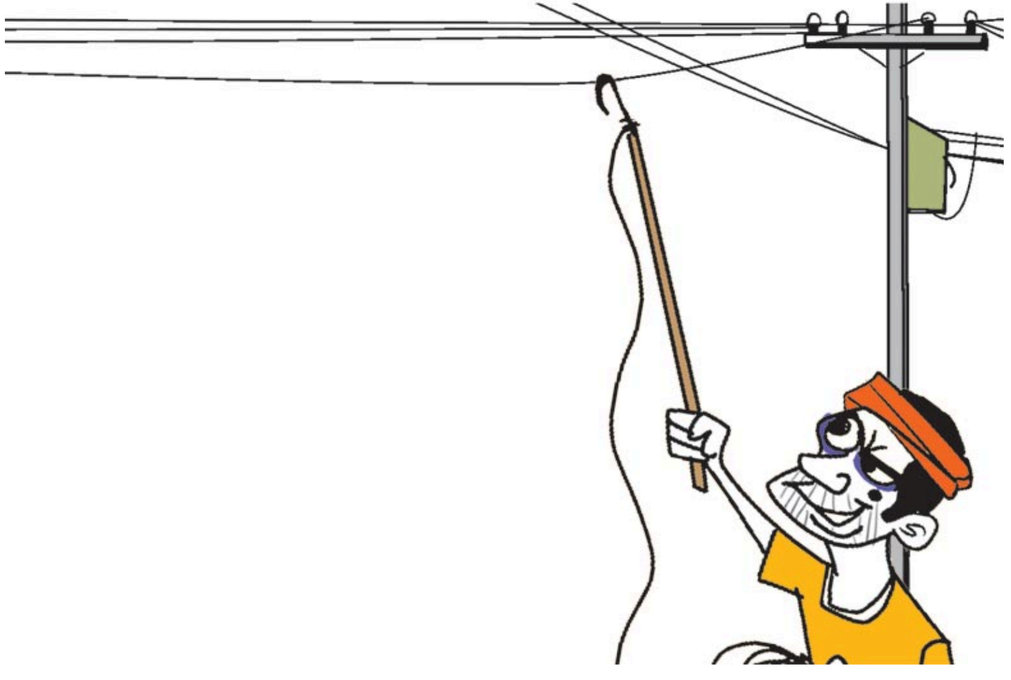


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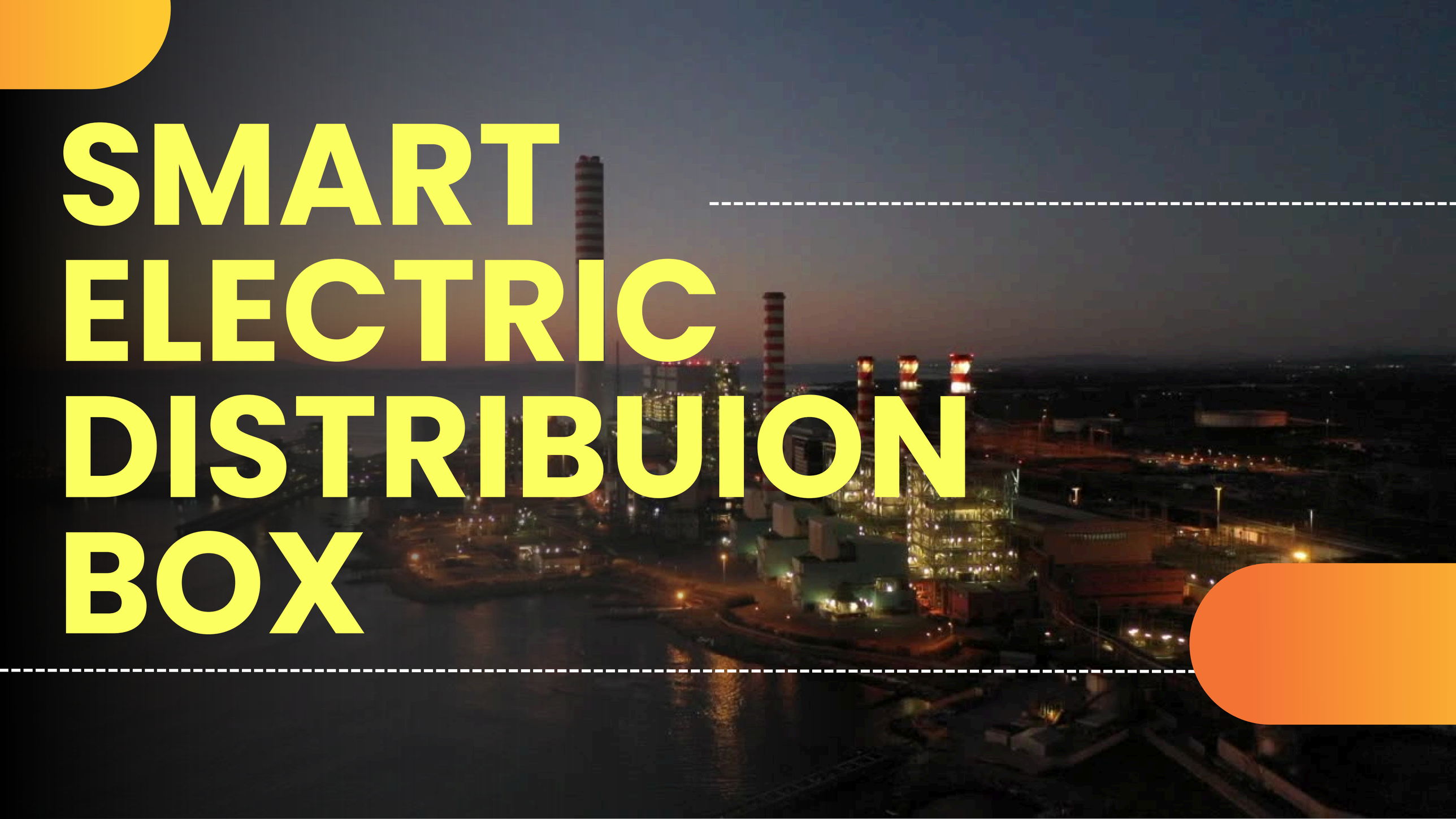
Two severe power outages affected most of northern and eastern India on 30 and 31 July 2012. The 30 July 2012 blackout affected over 400 million people and was briefly the largest power outage in history by number of people affected, beating the January 2001 blackout in Northern India (230 million affected).[1] The blackout on 31 July is the largest power outage in history. The outage affected more than 620 million people, about 9% of the world population,[2][3][4] or half of India's population, spread across 22 states in Northern, Eastern, and Northeast India.[5] An estimated 32 gigawatts of generating capacity was taken offline.[6] Of the affected population, 320 million initially had power, while the rest lacked direct access.[7] Electric service was restored in the affected locations between 31 July and 1 August 2012.[8][9]

India is the world's third largest producer and consumer of electricity after the United States and China; however, the electrical infrastructure was unreliable in those days.[10][11] The northern electrical grid had previously collapsed in 2001.[6] In 2012, an estimated 27% of energy generated was lost in transmission or stolen,[12] while peak supply fell short of demand by an average of 9%.[12] During that period, the nation suffered from frequent power outages that lasted as long as 10 hours[12] and about 25% of the population, about 300 million people, had no electricity at all.[12] Efforts are underway to reduce transmission and distribution losses and increase production.[13]





SMART ELECTRIC DISTRIBUION BOX



HOW CAN SMART DISTRIBUTION BOX SOLVE THIS PROBLEM

SMART DISTRIBUTION BOX

DIFFERENT TYPE OF SENSORS SUCH AS TEMPRATURE ,HUMIDITIY , CURRENT, EARTH LEAKAGE ETC

MICROCONTROLER

COMUNICATION SYSTEM

WIRED OF WIRE LESS



OUR SYSTEM COMPILE THE DATA COMEING FOR THE SENSORS OF SMART METER AND PRIDICT THE UPCOMING LOAD ON GRID .





IMPLIMENTAION OF SMART METERS IN AGREECULTURE IRRIGATION SECTOR

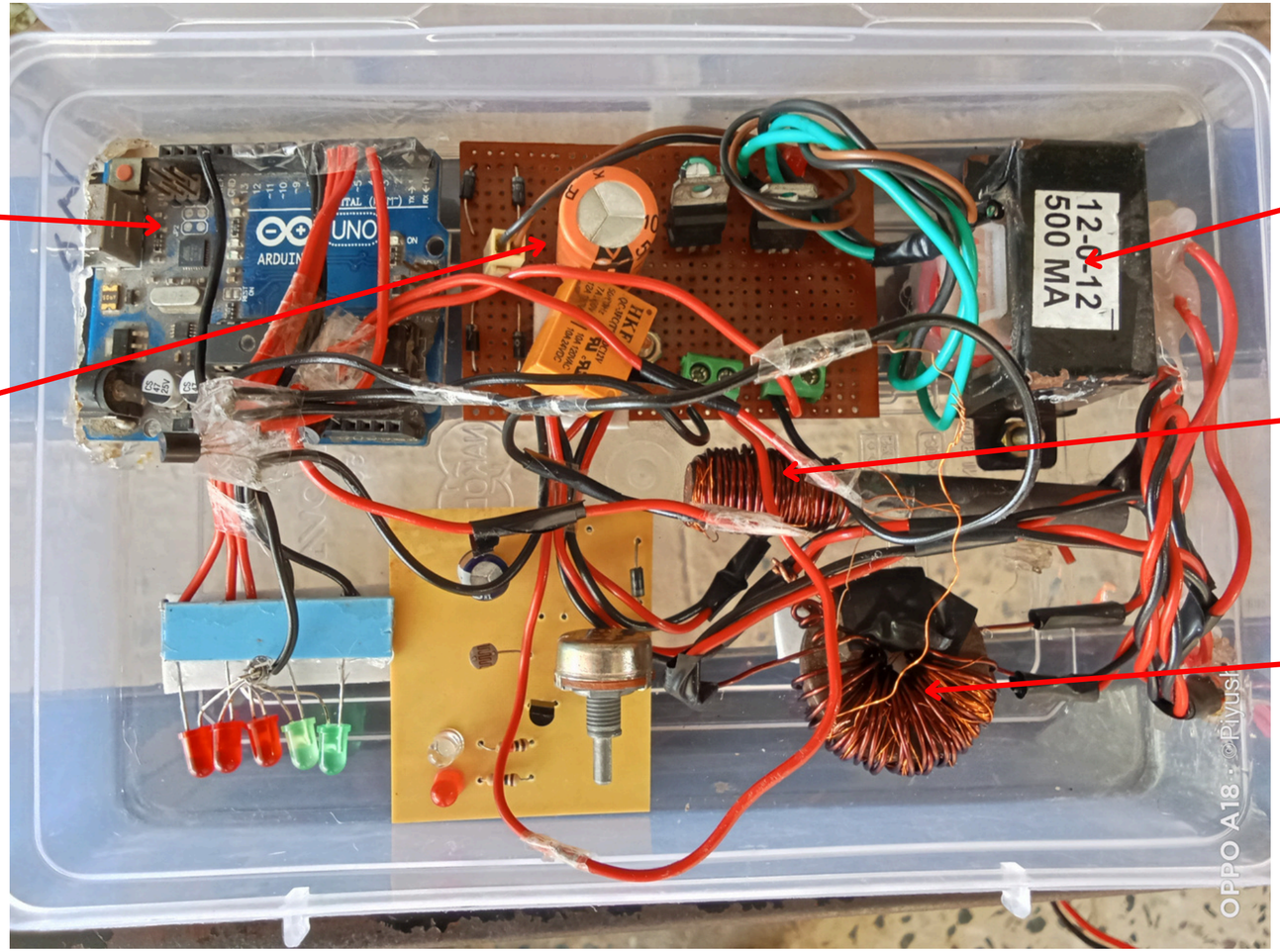




OUR SMART ELECTRIC DISTRIBUTION BOX

MICROCONTROLER

VOLTAGE REGULATOR CIRUIT

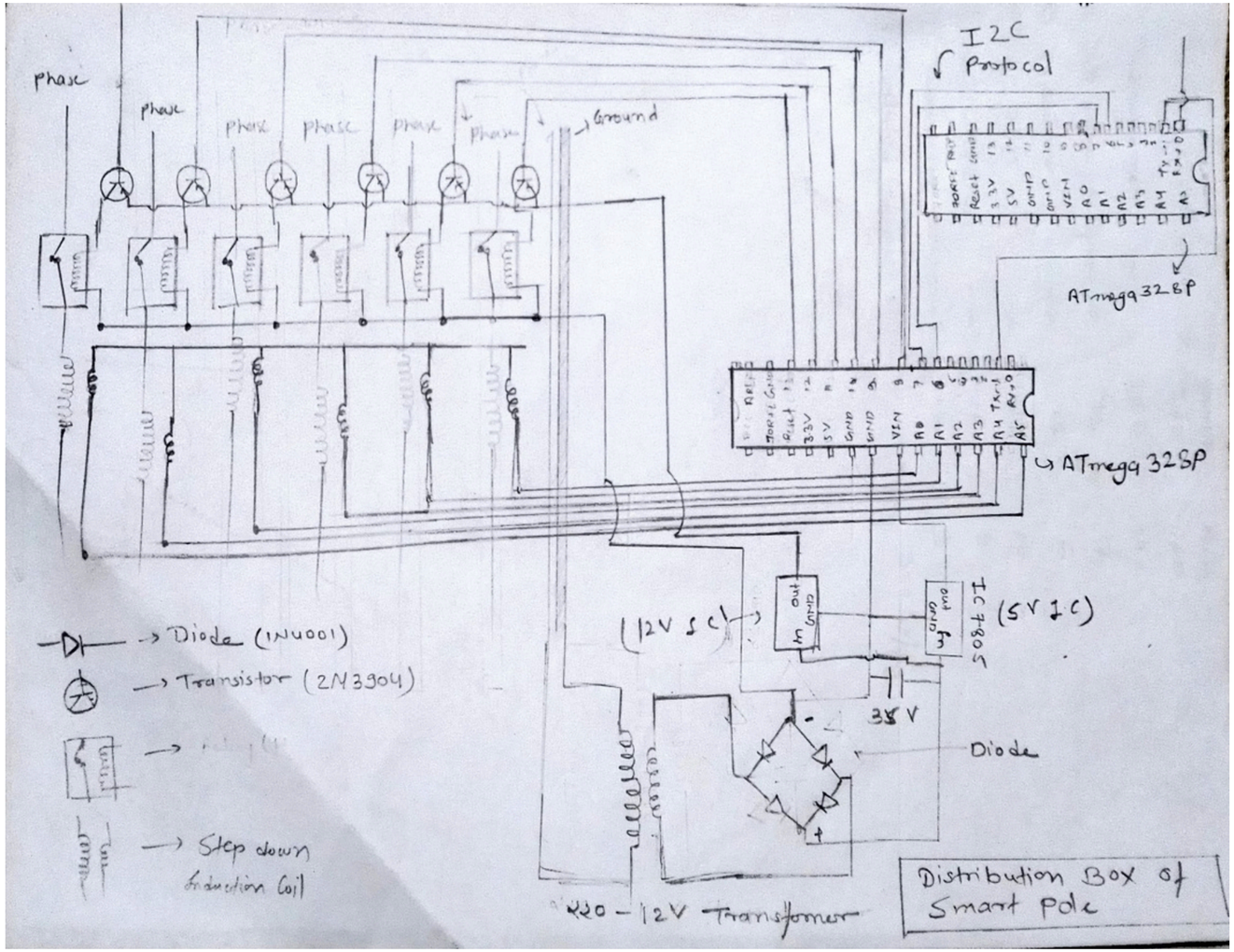


TRANSFORMER

CURRENT SENSOR

EARTH LEAKAGE SENSOR

BLOCK DIAGRAM OF SMART ELECTRIC DISTRIBUTION BOX




THANKYOU